

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Continuous Assignment

Submission:

Students are required to design, simulate, monitor, and document the following IoT-based systems using **Wokwi**, **VS Code**, and **Blynk Web Console**.

For each question, take clear screenshots of:

- Circuit design in Wokwi
- Complete source code (copy & paste)
- Serial Monitor output in VS Code
- Blynk Web Dashboard (widgets showing live data)

Combine all screenshots and source code for all questions into **one single PDF file** and submit it.

Questions:

1. Smart Parking Assistant

The Smart Parking Assistant system is designed to help drivers park vehicles safely by providing real-time distance feedback locally and remotely. An HC-SR04 ultrasonic sensor continuously measures the distance between the vehicle and an obstacle. Based on the detected distance, the system provides visual, audio, and cloud-based feedback.

Local Responses

- Distance < 10 cm
 - Fast continuous buzzer beeps
 - RGB LED turns RED (high risk)
- Distance between 10 cm and 30 cm
 - Slow buzzer beeps
 - RGB LED turns YELLOW (caution)
- Distance > 30 cm
 - Buzzer OFF
 - RGB LED turns GREEN (safe distance)

The measured distance is printed on the Serial Monitor every 500 ms.

Blynk Web Console

- Live distance value displayed using a Gauge widget
- Distance history visualized using a History Graph widget
- RGB LED status mirrored on Blynk using three LED widgets (Red, Yellow, Green) or one Styled Buttons.
- Parking status text shown as:
 - SAFE
 - CAUTION
 - DANGER

2. Intrusion Detection System

The Intrusion Detection System improves security by combining motion detection, RFID access control, and IoT-based alert monitoring. A PIR motion sensor detects human movement within a restricted area.

Intrusion Response

- When motion is detected:
 - Buzzer activates
 - Serial Monitor displays “Intrusion Alert!”

Authorized Access Control

- An MFRC522 RFID reader scans RFID cards
- If a valid RFID card is detected within 10 seconds of intrusion:
 - Relay (or LED) turns ON to simulate door unlocking
 - Blynk status updates to “Access Granted”
- If no valid card is detected:
 - Blynk displays “Access Denied”

Blynk Web Console

- Motion detection status shown using an LED widget
- System status displayed using a Label or Value Display widget (“Access Granted” / “Access Denied”)